Special Summer Integrated Learning Programme

Mathematics

Vectors in Two and Three Dimensions

Week 4 Day 1 PM

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Outline

Scalars versus vectors

In applications of mathematics, certain quantities are determined completely by their magnitude - for example, length, mass, area, temperature, and energy. Such a quantity is called a ***scalar***.

On the other hand, to describe the displacement of an object, two numbers are required: the magnitude and the direction of the displacement. To describe the velocity of a moving object, we must specify both the speed and the direction of travel. Quantities such as displacement, velocity, acceleration, and force that involve magnitude as well as direction are called directed quantities. One way to represent such quantities mathematically is through the use of ***vectors***.

Vectors in the coordinate plane

Intuitively, a ***vector*** in the coordinate plane is a line segment with an assigned direction (angle measured anti-clockwise from the positive x-axis), i.e. an arrow. Unlike numbers, we normally denote a vector by a bold letter $\boldsymbol{v}$ (more for printed text) or a letter with a small arrow above it $\vec{v}$.

Definition:

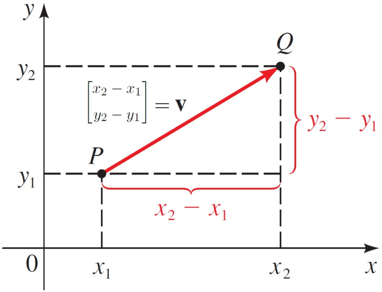
If a vector $\vec{v} = \overrightarrow{PQ}$ is represented in the plane with initial point $P(x_1, y_1)$ and terminal point (arrowhead) $Q(x_2, y_2)$, then we represent $\vec{v}$ as the column ordered pair

$$\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix},$$

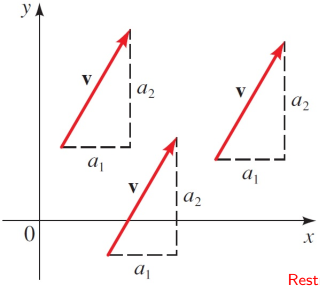
where $x = x_2 - x_1$ is the ***horizontal component*** of $\vec{v}$ and $y = y_2 - y_1$ is the ***vertical component*** of $\vec{v}$.

Alternative notations: We also represent $\vec{v}$ as $\langle x, y \rangle$ or $\big[x, y\big]$.

Vectors in the coordinate plane

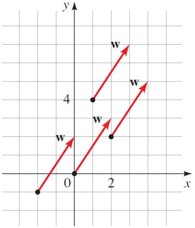


Remark: Note that a vector in the coordinate plane is defined by its two components, regardless of its position. Two vectors with the same horizontal and vertical components are equal.



Vectors in the coordinate plane - example

E.g.: $\vec{w} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ at different positions.



E.g.: The vector $\vec{u}$ has an initial point $(-2, 5)$ and terminal point $(3, 7)$, then

$$\vec{u} = \begin{bmatrix} 3-(-2) \\ 7-5 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

E.g.: The vector $\vec{v} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ has an initial point at $(2, 4)$. If its terminal point is (x, y), then

$$\begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} x-2 \\ y-4 \end{bmatrix} \Leftrightarrow \begin{cases} 3 = x-2 \\ 7 = y-4 \end{cases} \Leftrightarrow \begin{cases} x = 5 \\ y = 11. \end{cases}$$

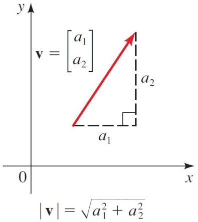
MAGNITUDE/LENGTH

DEFINITION:

The ***magnitude/length*** of a vector $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ is

$$||\vec{v}|| = \sqrt{x^2 + y^2}.$$

Remark: Some textbooks use $|\vec{v}|$ as the magnitude. The double vertical bars notation $||\vec{v}||$ is to distinguish it from the the absolute value of a number.



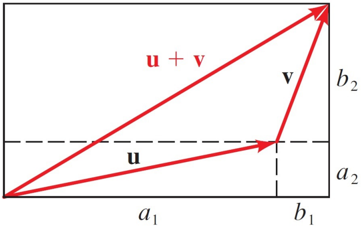
Vector addition

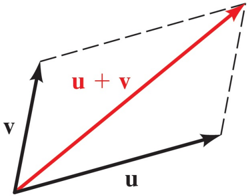
Definition:

For any $\vec{u} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, we define the vector addition

$$\vec{u} + \vec{v} := \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}.$$

Geometrically, $\vec{u} + \vec{v}$ is the main diagonal of the parallelogram formed by $\vec{u}$ and $\vec{v}$:





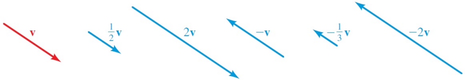
Scalar multiplication

Definition:

For any $\vec{v} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $c \in \mathbb{R}$, we define the scalar multiplication

$$c\vec{v} := \begin{bmatrix} ca_1 \\ ca_2 \end{bmatrix}.$$

Geometrically, $c\vec{v}$ corresponds to scaling $\vec{v}$ by a factor of c if $c \geq 0$; or flipping the direction and scaling $\vec{v}$ by a factor of $|c|$ if $c < 0$.



Notation: We denote $(-1)\vec{v}$ by $-\vec{v}$.

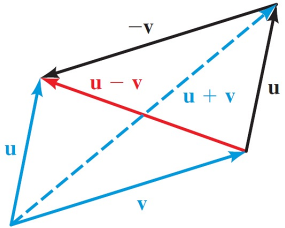
Vector subtraction

Definition:

For any $\vec{u} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, we define the vector subtraction

$$\vec{u} - \vec{v} := \vec{u} + (-\vec{v}) = \begin{bmatrix} a_1 - b_1 \\ a_2 - b_2 \end{bmatrix}.$$

Geometrically, $\vec{u} - \vec{v}$ is the anti-diagonal of the parallelogram formed by $\vec{u}$ and $\vec{v}$:



Operations on vectors - example

E.g.: If $\vec{u} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ then

$$\vec{u} + \vec{v} = \begin{bmatrix} 2 + (-1) \\ -3 + 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{u} - \vec{v} = \begin{bmatrix} 2 - (-1) \\ -3 - 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$$2\vec{u} = \begin{bmatrix} 2 \cdot 2 \\ 2 \cdot (-3) \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \end{bmatrix}$$

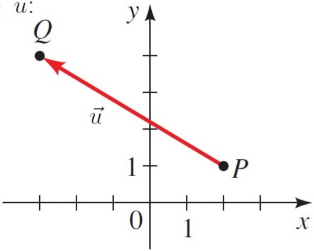
$$-\vec{v} = \begin{bmatrix} -(-1) \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$2\vec{u} + 3\vec{v} = 2\begin{bmatrix} 2 \\ -3 \end{bmatrix} + 3\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 + 3(-1) \\ 2(-3) + 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

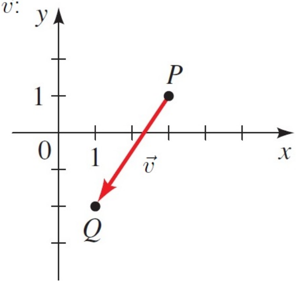
Activity 1

(a) Find the following vectors with initial point P and terminal point Q. Compute their lengths also.

(i) $\vec{u}$:



(ii) $\vec{v}$:



(iii) $\vec{w}$: $P(-1,3)$, $Q(-6,-1)$.

(iv) $\vec{x}$: $P(5,3)$, $Q(1,0)$.

(b) Using the vectors from part (a), compute and sketch

(i) $\vec{u}+\vec{v}$.

(ii) $\vec{u}-\vec{v}$.

(iii) $\vec{v}-\vec{u}$.

(iv) $2\vec{u}-3\vec{x}$.

Activity 1: solution

(a) (i) $P(2,1)$, $Q(-3,4)$, so $\vec{u} = \begin{bmatrix} -3-2 \\ 4-1 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$.

$||\vec{u}|| = \sqrt{(-5)^2 + 3^2} = \sqrt{34}$.

(ii) $P(3,1)$, $Q(1,-2)$, so $\vec{v} = \begin{bmatrix} 1-3 \\ -2-1 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$.

$||\vec{v}|| = \sqrt{(-2)^2 + (-3)^2} = \sqrt{13}$.

(iii) $\vec{w} = \begin{bmatrix} -6-(-1) \\ -1-3 \end{bmatrix} = \begin{bmatrix} -5 \\ -4 \end{bmatrix}$.

$||\vec{w}|| = \sqrt{(-5)^2 + (-4)^2} = \sqrt{41}$.

(iv) $\vec{x} = \begin{bmatrix} 1-5 \\ 0-3 \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$. $||\vec{x}|| = \sqrt{(-4)^2 + (-3)^2} = 5$.

Activity 1: solution

(b) $\vec{u} = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$.

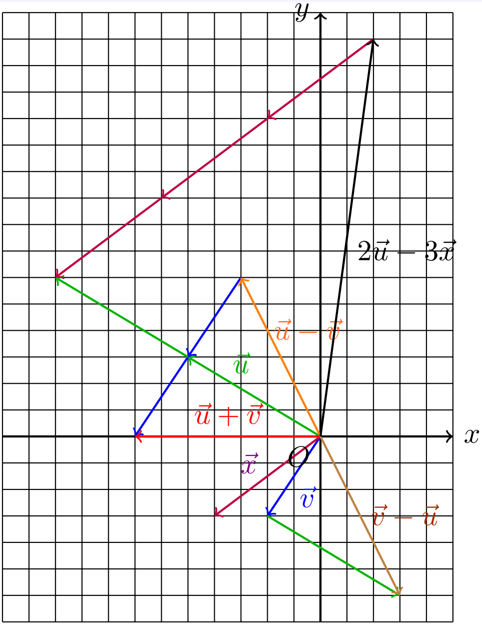
(i) $\vec{u} + \vec{v} = \begin{bmatrix} -5 + (-2) \\ 3 + (-3) \end{bmatrix} = \begin{bmatrix} -7 \\ 0 \end{bmatrix}$.

(ii) $\vec{u} - \vec{v} = \begin{bmatrix} -5 - (-2) \\ 3 - (-3) \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$.

(iii) $\vec{v} - \vec{u} = \begin{bmatrix} (-2) - (-5) \\ (-3) - 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}$.

(iv) $2\vec{u} - 3\vec{x} = \begin{bmatrix} 2(-5) - 3(-4) \\ 2(3) - 3(-3) \end{bmatrix} = \begin{bmatrix} 2 \\ 15 \end{bmatrix}$.

Activity 1: solution



Properties of vectors

The following properties can be proved easily from the definitions. The ***zero vector*** is the vector $\vec{0} := \langle 0, 0 \rangle$. It plays the same role for addition of vectors as the number 0 does for addition of real numbers.

PROPERTIES OF VECTORS

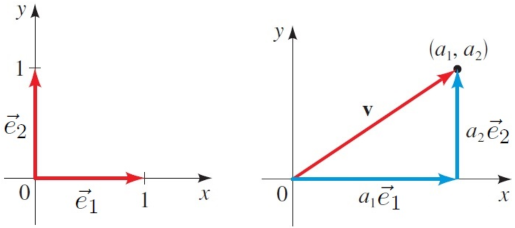
Vector addition	Multiplication by a scalar
$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$	$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$	$(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
$\mathbf{u} + \mathbf{0} = \mathbf{u}$	$(cd)\mathbf{u} = c(d\mathbf{u}) = d(c\mathbf{u})$
$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$	$1\mathbf{u} = \mathbf{u}$
Length of a vector	$0\mathbf{u} = \mathbf{0}$
$\lvert c\mathbf{u} \rvert = \lvert c \rvert \lvert \mathbf{u} \rvert$	$c\mathbf{0} = \mathbf{0}$

Remark: In fact, mathematicians define vectors to be any objects that satisfy all the properties above, generalizing the notion of vector beyond arrows. Under this definition, sequences, polynomials, matrices, etc. are vectors.

Unit vector

A vector of length 1 is called a ***unit vector***. Two useful unit vectors are $\vec{e}_1$ and $\vec{e}_2$, defined by

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$



As you can see from the right figure above, any vector can be expressed in terms of $\vec{e}_1$ and $\vec{e}_2$ by

$$\vec{v} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = a_1\vec{e}_1 + a_2\vec{e}_2.$$

Remark: In physics, $\vec{e}_1$ is denoted by $\hat{\boldsymbol{\imath}}$, $\vec{e}_2$ is denoted by $\hat{\boldsymbol{\jmath}}$.

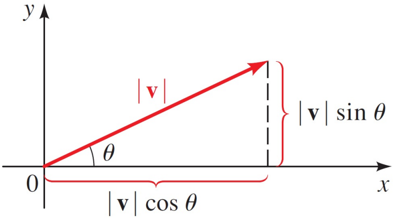
Components of a vector

Remark: Note that given any non-zero vector $\vec{u}$, the vector $\frac{1}{||\vec{u}||}\vec{u}$ is a unit vector pointing in the same direction as $\vec{u}$.

Relationships between magnitude, direction and the components of a vector

Let $\vec{v}$ be a vector with magnitude $||\vec{v}||$ and direction θ, then we can show by trigonometry that

$$\vec{v} = \begin{bmatrix} ||\vec{v}|| \cos\theta \\ ||\vec{v}|| \sin\theta \end{bmatrix} = (||\vec{v}|| \cos\theta)\, \vec{e}_1 + (||\vec{v}|| \sin\theta)\, \vec{e}_2.$$

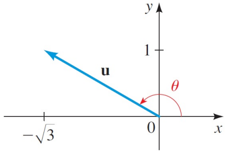


Components of a vector - example

E.g.: A vector $\vec{v}$ has length 8 and direction $\frac{\pi}{3}$. Then

$$\vec{v} = \begin{bmatrix} 8\cos\frac{\pi}{3} \\ 8\sin\frac{\pi}{3} \end{bmatrix} = \begin{bmatrix} 8\cdot\frac{1}{2} \\ 8\cdot\frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 4 \\ 4\sqrt{3} \end{bmatrix}.$$

E.g.: Given $\vec{u} = -\sqrt{3}\vec{e}_1 + \vec{e}_2$, then its direction θ is in the 2nd quadrant



So

$$\tan\theta = \frac{1}{-\sqrt{3}} \Rightarrow \theta = \frac{5\pi}{6}$$

and

$$||\vec{u}|| = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = 2.$$

Activity 2

(a) Find $||\vec{u}||, ||\vec{v}||, ||-2\vec{u}||, \left|\left|\frac{1}{2}\vec{v}\right|\right|, ||3\vec{u}-4\vec{v}||$ and express $3\vec{u}-4\vec{v}$ in terms of $\vec{e}_1$ and $\vec{e}_2$ for the given $\vec{u}$ and $\vec{v}$.

(i) $\vec{u} = \begin{bmatrix} 2 \\ 7 \end{bmatrix}, \vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$

(ii) $\vec{u} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}, \vec{v} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}.$

(b) Express the following vectors in terms of $\vec{e}_1$ and $\vec{e}_2$ with the given length and direction.

(i) $||\vec{w}|| = 4,\ \theta = \frac{5\pi}{6}.$

(ii) $||\vec{x}|| = \sqrt{3},\ \theta = 300^\circ.$

(c) Find the length and direction of the following vectors.

(i) $\vec{y} = -\frac{\sqrt{2}}{2}\vec{e}_1 - \frac{\sqrt{2}}{2}\vec{e}_2.$

(ii) $\vec{z} = \begin{bmatrix} -12 \\ 5 \end{bmatrix}.$

Activity 2: solution

(a) (i) $3\vec{u} - 4\vec{v} = (3(2) - 4(3))\vec{e}_1 + (3(7) - 4(1))\vec{e}_2 = -6\vec{e}_1 + 17\vec{e}_2.$

$$||\vec{u}|| = \sqrt{2^2 + 7^2} = \sqrt{53}, \qquad ||\vec{v}|| = \sqrt{3^2 + 1^2} = \sqrt{10},$$

$$||-2\vec{u}|| = 2||\vec{u}|| = 2\sqrt{53}, \qquad \left|\left|\frac{1}{2}\vec{v}\right|\right| = \frac{1}{2}||\vec{v}|| = \frac{\sqrt{10}}{2},$$

$$||3\vec{u} - 4\vec{v}|| = \sqrt{(-6)^2 + 17^2} = \sqrt{325}.$$

(ii) $3\vec{u} - 4\vec{v} = (3(-2) - 4(2))\vec{e}_1 + (3(5) - 4(-8))\vec{e}_2 = -14\vec{e}_1 + 47\vec{e}_2.$

$$||\vec{u}|| = \sqrt{(-2)^2 + 5^2} = \sqrt{29}, \qquad ||\vec{v}|| = \sqrt{2^2 + (-8)^2} = \sqrt{68},$$

$$||-2\vec{u}|| = 2||\vec{u}|| = 2\sqrt{29}, \qquad \left|\left|\frac{1}{2}\vec{v}\right|\right| = \frac{1}{2}||\vec{v}|| = \frac{\sqrt{68}}{2},$$

$$||3\vec{u} - 4\vec{v}|| = \sqrt{(-14)^2 + 47^2} = \sqrt{2405}.$$

Activity 2: solution

(b) (i)

$$\begin{aligned}
\vec{w} &= (||\vec{w}||\cos\theta)\,\vec{e}_1 + (||\vec{w}||\sin\theta)\,\vec{e}_2 \\
&= \left(4\cos\frac{5\pi}{6}\right)\vec{e}_1 + \left(4\sin\frac{5\pi}{6}\right)\vec{e}_2 \\
&= \left(4\left(-\frac{\sqrt{3}}{2}\right)\right)\vec{e}_1 + \left(4\cdot\frac{1}{2}\right)\vec{e}_2 \\
&= -2\sqrt{3}\vec{e}_1 + 2\vec{e}_2.
\end{aligned}$$

(ii)

$$\begin{aligned}
\vec{x} &= (||\vec{x}||\cos\theta)\,\vec{e}_1 + (||\vec{x}||\sin\theta)\,\vec{e}_2 \\
&= \left(\sqrt{3}\cos 300^\circ\right)\vec{e}_1 + \left(\sqrt{3}\sin 300^\circ\right)\vec{e}_2 \\
&= \left(\sqrt{3}\cdot\frac{1}{2}\right)\vec{e}_1 + \left(\sqrt{3}\left(-\frac{\sqrt{3}}{2}\right)\right)\vec{e}_2 \\
&= \frac{\sqrt{3}}{2}\vec{e}_1 - \frac{3}{2}\vec{e}_2.
\end{aligned}$$

Activity 2: solution

(c) (i) $||\vec{y}|| = \sqrt{\left(-\frac{\sqrt{2}}{2}\right)^2 + \left(-\frac{\sqrt{2}}{2}\right)^2} = \sqrt{\frac{2}{4} + \frac{2}{4}} = 1$. If the direction of $\vec{y}$ is θ, then θ is in the 3rd quadrant because both the horizontal and vertical components of $\vec{y}$ are negative. Thus

$$\tan\theta = \frac{-\frac{\sqrt{2}}{2}}{-\frac{\sqrt{2}}{2}} = 1 \;\Rightarrow\; \theta = \pi + \arctan(1) = \frac{5\pi}{4}.$$

(ii) $||\vec{z}|| = \sqrt{(-12)^2 + 5^2} = \sqrt{\frac{2}{4} + \frac{2}{4}} = \sqrt{169} = 13$. If the direction of $\vec{z}$ is θ, then θ is in the 2nd quadrant because its horizontal component is negative while its vertical component is positive. Thus

$$\tan\theta = \frac{5}{-12} \;\Rightarrow\; \theta = \pi + \arctan\left(\frac{5}{-12}\right) \approx 2.7468\,\mathrm{rad}\,.$$

Outline

1. Vectors in Two Dimensions
2. The Dot Product
3. Summary

The dot product for 2D vectors

Definition:

If $\vec{u} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, then their ***dot product***, denoted by $\vec{u} \cdot \vec{v}$, is the real number defined by

$$\vec{u} \cdot \vec{v} := a_1 b_1 + a_2 b_2.$$

E.g.: If $\vec{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, then

$$\vec{u} \cdot \vec{v} = (3)(4) + (-2)(5) = 2.$$

E.g.: If $\vec{u} = 2\vec{e}_1 + \vec{e}_2$ and $\vec{v} = 5\vec{e}_1 - 6\vec{e}_2$, then

$$\vec{u} \cdot \vec{v} = (2)(5) + (1)(-6) = 4.$$

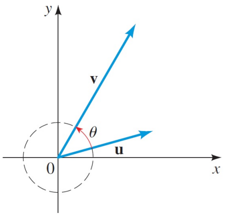
Properties of the dot product

By using the definition of the dot product, we can easily prove the following properties.

Properties of the Dot Product

- $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$.
- $(c\vec{u}) \cdot \vec{v} = c\,(\vec{u} \cdot \vec{v}) = u \cdot (c\vec{v})$.
- $(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$.
- $\vec{u} \cdot \vec{u} = ||\vec{u}||^2$.

So far, the dot product is defined algebraically without any geometrical meaning. It turns out that the dot product of $\vec{u}$ and $\vec{v}$ is related to the ***angle between the two vectors*** (only defined when $\vec{u}, \vec{v}$ are non-zero), which is defined to be the smaller angle $\theta \in [0, \pi]$ formed by $\vec{u}$ and $\vec{v}$:



The dot product theorem

The Dot Product Theorem

If θ is the angle between two non-zero vectors $\vec{u}$ and $\vec{v}$, then

$$\vec{u} \cdot \vec{v} = ||\vec{u}||||\vec{v}|| \cos \theta.$$

Please refer to page 640 of the textbook for the proof. The proof applies the Law of Cosines.

E.g.: If θ is the angle between $\vec{u} = 2\vec{e}_1 + 5\vec{e}_2$ and $\vec{v} = 4\vec{e}_1 - 3\vec{e}_2$, then

$$\begin{aligned} & \vec{u} \cdot \vec{v} = ||\vec{u}||||\vec{v}|| \cos \theta \\ \Leftrightarrow \quad & (2)(4) + (5)(-3) = \sqrt{2^2 + 5^2}\sqrt{4^2 + (-3)^2} \cos \theta \\ \Leftrightarrow \quad & \cos \theta = \frac{-7}{5\sqrt{29}} \\ \Leftrightarrow \quad & \theta = \arccos\left(\frac{-7}{5\sqrt{29}}\right) \approx 1.8338 \, \mathrm{rad}\,. \end{aligned}$$

Orthogonal vectors

The dot product theorem is useful for finding the angle between two non-zero vectors, in particular it can easily tell us when are the two vectors perpendicular to each other. Since $\cos\frac{\pi}{2} = 0$, so two non-zero vectors $\vec{u}$ and $\vec{v}$ are perpendicular to each other if

$$\vec{u}\cdot\vec{v} = ||\vec{u}||||\vec{v}||\cos\frac{\pi}{2} = 0.$$

Definition:

Two vectors $\vec{u}$ and $\vec{v}$ are ***orthogonal***/***perpendicular*** iff. $\vec{u}\cdot\vec{v} = 0$.

Remark: Note that the definition above extends the notion of perpendicularity to zero vector. If the two vectors are non-zero, then indeed $\vec{u}\cdot\vec{v} = 0$ iff. the angle between them is $\frac{\pi}{2}$. If one of both of the vectors are zero vector, then the angle between them is not defined. However, according to the definition above, the zero vector is perpendicular to any other vector.

Orthogonal vectors - example

E.g.: Without sketching the vectors $\vec{u} = 3\vec{e}_1 + 5\vec{e}_2$ and $\vec{v} = 2\vec{e}_1 - 8\vec{e}_2$, we can tell that they are not perpendicular to each other because

$$\vec{u} \cdot \vec{v} = (3)(2) + (5)(-8) = -34 \neq 0.$$

E.g.: The vectors $\vec{u} = 2\hat{\boldsymbol{i}} + \hat{\boldsymbol{j}}$ and $\vec{v} = \hat{\boldsymbol{i}} - 2\hat{\boldsymbol{j}}$ are perpendicular to each other because

$$\vec{u} \cdot \vec{v} = (2)(1) + (1)(-2) = 0.$$

Remark: In linear algebra, we'll extend vectors to dimension 3, 4, 5, and so on. It's impossible to visualize vectors in dimensions higher than 3, but thanks to the algebraic definitions of angle and orthogonality, we can still compute the angle between two vectors in high dimensions and check whether they are orthogonal or not.

Activity 3

(a) Find $\vec{u} \cdot \vec{v}$ and the angle between $\vec{u}, \vec{v}$ for the following vectors.

(i) $\vec{u} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

(ii) $\vec{u} = \vec{e}_1 + \sqrt{3}\vec{e}_2$, $\vec{v} = -\sqrt{3}\vec{e}_1 + \vec{e}_2$.

(iii) $\vec{u} = -5\hat{\boldsymbol{j}}$, $\vec{v} = -\hat{\boldsymbol{i}} - \sqrt{3}\hat{\boldsymbol{j}}$.

(b) Which of the following pairs of vectors are orthogonal to each other?

(i) $\vec{u} = 6\vec{e}_1 + 4\vec{e}_2$, $\vec{v} = -2\vec{e}_1 + 3\vec{e}_2$.

(ii) $\vec{u} = -2\hat{\boldsymbol{i}} + 6\hat{\boldsymbol{j}}$, $\vec{v} = 4\hat{\boldsymbol{i}} + 2\hat{\boldsymbol{j}}$.

(c) Given $\vec{x} = 2\vec{e}_1 + \vec{e}_2$, $\vec{y} = \vec{e}_1 - 3\vec{e}_2$ and $\vec{z} = 3\vec{e}_1 + 4\vec{e}_2$, find

(i) $\vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{z}$.

(ii) $(\vec{x} \cdot \vec{y})(\vec{x} \cdot \vec{z})$.

(d) Simplify $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b})$.

Activity 3: solution

(a) (i) $\vec{u}\cdot\vec{v} = (2)(1) + (0)(1) = 2.$

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{||\vec{u}||||\vec{v}||} = \frac{2}{\sqrt{2^2+0^2}\sqrt{1^2+1^2}} = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\Rightarrow \theta = \frac{\pi}{4}.$$

(ii) $\vec{u}\cdot\vec{v} = (1)(-\sqrt{3}) + (\sqrt{3})(1) = 0.$

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{||\vec{u}||||\vec{v}||} = 0 \;\Rightarrow\; \theta = \frac{\pi}{2}.$$

(iii) $\vec{u}\cdot\vec{v} = (0)(-1) + (-5)(-\sqrt{3}) = 5\sqrt{3}.$

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{||\vec{u}||||\vec{v}||} = \frac{5\sqrt{3}}{\sqrt{0^2+(-5)^2}\sqrt{(-1)^2+(-\sqrt{3})^2}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}.$$

Activity 3: solution

(b) (i) $\vec{u} \cdot \vec{v} = (6)(-2) + (4)(3) = 0$, hence they are orthogonal to each other.

(ii) $\vec{u} \cdot \vec{v} = (-2)(4) + (6)(2) = 4 \neq 0$, hence they are not orthogonal to each other.

(c) Given $\vec{x} = 2\vec{e}_1 + \vec{e}_2$, $\vec{y} = \vec{e}_1 - 3\vec{e}_2$ and $\vec{z} = 3\vec{e}_1 + 4\vec{e}_2$, find

(i) $\vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{z} = ((2)(1) + (1)(-3)) + ((2)(3) + (1)(4)) = 9.$

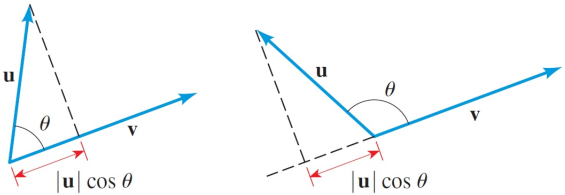
(ii) $(\vec{x} \cdot \vec{y})(\vec{x} \cdot \vec{z}) = ((2)(1) + (1)(-3))((2)(3) + (1)(4)) = (-1)(10) = -10.$

(d)

$$\begin{aligned}(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) &= \vec{a} \cdot (\vec{a} - \vec{b}) + \vec{b} \cdot (\vec{a} - \vec{b}) \\ &= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} - \vec{b} \cdot \vec{b} \\ &= ||\vec{a}||^2 - ||\vec{b}||^2.\end{aligned}$$

The component of $\vec{u}$ along $\vec{v}$

Intuitively, the component of $\vec{u}$ along $\vec{v}$ is the magnitude of the portion of $\vec{u}$ that points in the direction of $\vec{v}$. Notice that the component of $\vec{u}$ along $\vec{v}$ is negative if the angle between the two vectors is $> \frac{\pi}{2}$.



With the sign taken into consideration, trigonometry gives the following definition.

Definition:

The ***component of $\vec{u}$ along/in the direction of $\vec{v}$*** is the real number

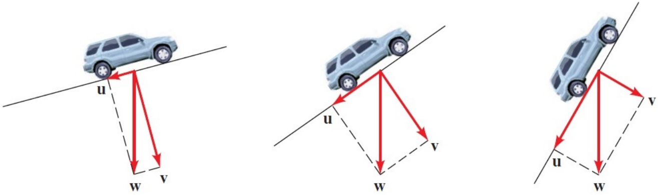
$$\text{comp}_{\vec{v}}(\vec{u}) := ||\vec{u}|| \cos\theta = \frac{\vec{u} \cdot \vec{v}}{||\vec{v}||}.$$

The component of $\vec{u}$ along $\vec{v}$ in physics

In analyzing forces in physics and engineering, it's often helpful to express a vector as a sum of two vectors lying in perpendicular directions. For example, suppose a car is parked on an inclined driveway. The weight of the car is a vector $\vec{w}$ that points directly downward. We can write

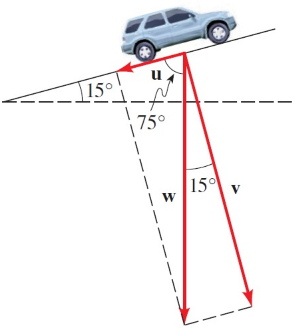
$$\vec{w} = \vec{u} + \vec{v}$$

where $\vec{u}$ is parallel to the driveway and $\vec{v}$ is perpendicular to the driveway. The vector $\vec{u}$ is the force that tends to roll the car down the driveway, and $\vec{v}$ is the force experienced by the surface of the driveway. The magnitudes of these forces are the components of $\vec{w}$ along $\vec{u}$ and $\vec{v}$, respectively.



THE COMPONENT OF $\vec{u}$ ALONG $\vec{v}$ - EXAMPLE

E.g.: A car weighing 3000 lb is parked on a driveway that is inclined 15° to the horizontal. Then the magnitude of the force (resistance between the tyres and the driveway) required to prevent the car from rolling down the driveway is the component of the weight along a vector $\vec{u}$ parallel to the driveway.



$$\text{comp}_{\vec{u}}(\vec{w}) = ||\vec{w}|| \cos 75^\circ \approx 776 \text{ lb.}$$

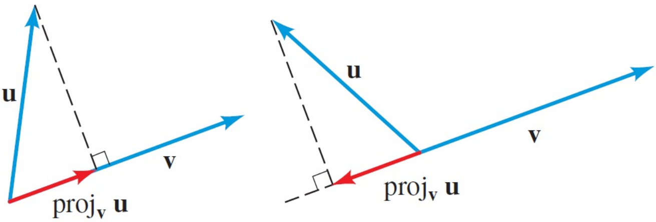
E.g.: Given $\vec{u} = \vec{e}_1 + 4\vec{e}_2$ and $\vec{v} = -2\vec{e}_1 + \vec{e}_2$, then

$$\text{comp}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{||\vec{v}||} = \frac{(1)(-2) + (4)(1)}{\sqrt{(-2)^2 + 1^2}} = \frac{2}{\sqrt{5}},$$

$$\text{comp}_{\vec{u}}(\vec{v}) = \frac{\vec{v} \cdot \vec{u}}{||\vec{u}||} = \frac{(-2)(1) + (1)(4)}{\sqrt{1^2 + 4^2}} = \frac{2}{\sqrt{17}}.$$

The projection of $\vec{u}$ onto $\vec{v}$

The component of $\vec{u}$ along $\vec{v}$ only gives us a scalar, but sometimes we need the vector parallel to $\vec{v}$ whose length is the component of $\vec{u}$ along $\vec{v}$, which is the red vectors indicated below.



Such vector is called the projection of $\vec{u}$ onto $\vec{v}$. Geometrically, it's the shadow of $\vec{u}$ that lies on $\vec{v}$ if we shine light perpendicular to $\vec{v}$. Projection of vectors is very useful in physics. For instance, if we refer back to the car example, the vectors $\vec{u}$ and $\vec{v}$ in decomposition

$$\vec{w} = \vec{u} + \vec{v}$$

are projections in two perpendicular directions.

THE PROJECTION OF $\vec{u}$ ONTO $\vec{v}$

Since the projection of $\vec{u}$ onto $\vec{v}$ has length $\text{comp}_{\vec{v}}(\vec{u})$ and is parallel to $\vec{v}$, it's thus given by the unit vector pointing in the direction of $\vec{v}$, i.e. $\frac{1}{||\vec{v}||}\vec{v}$, scaled by $\text{comp}_{\vec{v}}(\vec{u})$. This gives rise to the following definition.

DEFINITION:

The ***projection of $\vec{u}$ onto $\vec{v}$*** is the vector

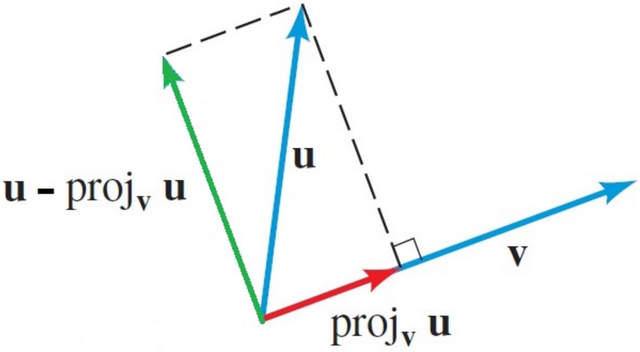
$$\text{proj}_{\vec{v}}(\vec{u}) := \text{comp}_{\vec{v}}(\vec{u})\left(\frac{1}{||\vec{v}||}\vec{v}\right) = \frac{\vec{u}\cdot\vec{v}}{||\vec{v}||^2}\vec{v}.$$

Remark: Note that

$$\vec{u} = \text{proj}_{\vec{v}}(\vec{u}) + (\vec{u} - \text{proj}_{\vec{v}}(\vec{u})),$$

where $\text{proj}_{\vec{v}}(\vec{u})$ is parallel to $\vec{v}$, $\vec{u} - \text{proj}_{\vec{v}}(\vec{u})$ is perpendicular to $\vec{v}$.

The projection of $\vec{u}$ onto $\vec{v}$

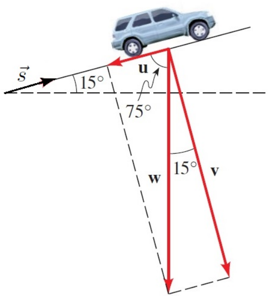


The projection of $\vec{u}$ onto $\vec{v}$ - example

E.g.: Continuing from the previous car example, we can decompose the car's weight as $-3000\vec{e}_2 = \vec{w} = \vec{u} + \vec{v}$, where

$$\vec{u} = \text{proj}_{\vec{s}}(\vec{w}), \qquad \vec{v} = \vec{w} - \text{proj}_{\vec{s}}(\vec{w}),$$

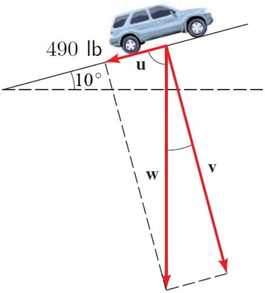
and $\vec{s} = (\cos 15°)\vec{e}_1 + (\sin 15°)\vec{e}_2$ is a unit vector parallel to the inclined driveway. So



$$\vec{u} = \text{proj}_{\vec{s}}(\vec{w}) = \frac{\vec{w}\cdot\vec{s}}{||\vec{s}||^2}\vec{s} = \frac{(0)(\cos 15°) + (-3000)(\sin 15°)}{1^2}\vec{s}$$
$$= -3000\left((\sin 15° \cos 15°)\vec{e}_1 + (\sin^2 15°)\vec{e}_2\right)$$
$$\vec{v} = \vec{w} - \text{proj}_{\vec{s}}(\vec{w})$$
$$= -3000\vec{e}_2 - (-3000)\left((\sin 15° \cos 15°)\vec{e}_1 + (\sin^2 15°)\vec{e}_2\right)$$
$$= 3000\left((\sin 15° \cos 15°)\vec{e}_1 - (\cos^2 15°)\vec{e}_2\right).$$

Activity 4

(a) A car is on a driveway that is inclined 10° to the horizontal. A force of 490 lb is required to keep the car from rolling down the driveway.



(i) Find the weight of the car.

(ii) Find the magnitude of the force that the car exerts against the driveway perpendicularly.

(b) Given $\vec{u} = 3\vec{e}_1 + 4\vec{e}_2$ and $\vec{w} = -\vec{e}_1 + \vec{e}_2$. Write $\vec{w} = \vec{x} + \vec{y}$, where $\vec{x}$ is parallel to $\vec{u}$ and $\vec{y}$ is perpendicular to $\vec{u}$. Verify your answers by checking $\vec{x}$ and $\vec{y}$ are perpendicular.

Activity 4: solution

(a) (i) Let the weight be $\vec{w}$, then $||\vec{u}|| = 490$ is the component of $\vec{w}$ along $\vec{u}$, i.e.

$$490 = ||\vec{w}|| \cos 80^\circ \;\Rightarrow\; ||\vec{w}|| = 490 \sec 80^\circ = 2821.8 \text{ lb.}$$

So $\vec{w} = -2821.8\hat{\boldsymbol{j}}$.

(ii) That is $||\vec{v}||$, the component of $\vec{w}$ along $\vec{v}$, which is given by

$$||\vec{w}|| \sin 80^\circ = 2821.8 \sin 80^\circ = 2778.9 \text{ lb.}$$

(b)

$$\begin{aligned}
\vec{x} = \mathsf{proj}_{\vec{u}}(\vec{w}) &= \frac{\vec{w}\cdot\vec{u}}{||\vec{u}||^2}\vec{u} \\
&= \frac{(-1)(3)+(1)(4)}{3^2+4^2}\,(3\vec{e}_1 + 4\vec{e}_2) \\
&= \frac{3}{25}\vec{e}_1 + \frac{4}{25}\vec{e}_2.
\end{aligned}$$

Activity 4: solution

(b)

$$\begin{aligned}\vec{y} &= \vec{w} - \mathsf{proj}_{\vec{u}}(\vec{w}) \\ &= -\vec{e}_1 + \vec{e}_2 - \left(\frac{3}{25}\vec{e}_1 + \frac{4}{25}\vec{e}_2\right) \\ &= -\frac{28}{25}\vec{e}_1 + \frac{21}{25}\vec{e}_2.\end{aligned}$$

We verify that

$$\vec{x} \cdot \vec{y} = \left(\frac{3}{25}\right)\left(-\frac{28}{25}\right) + \left(\frac{4}{25}\right)\left(\frac{21}{25}\right) = 0.$$

Hence, $\vec{x}$ and $\vec{y}$ are indeed perpendicular to each other.

Outline

Summary

- Vectors in Two Dimensions
  - scalars vs. vectors.
  - vectors in the coordinate plane.
  - the magnitude/length of a vector.
  - vector addition, subtraction and scalar multiplication.
  - properties of vectors, unit vectors, $\vec{e}_1, \vec{e}_2$, components of a vector
- The Dot Product
  - the dot product for 2D vectors.
  - properties of the dot product.
  - angle between two vectors and the dot product theorem.
  - orthogonal vectors.
  - the component of $\vec{u}$ along $\vec{v}$.
  - the projection of $\vec{u}$ on $\vec{v}$.

Resources

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 9.1, 9.2.